

Enhancing Robustness in the Wild: Inference-Time Generative Restoration for Occluded Visual Recognition in Affective Computing

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Abstract - Deep vision models achieve high accuracy in controlled environments but exhibit significant performance degradation in real-world settings, where partial obstructions and noise frequently obscure critical data. Conventional robustness strategies either train classifiers to tolerate missing information, thereby discarding latent critical signals, or employ Generative Adversarial Networks (GANs) exclusively for offline dataset augmentation. Neither approach actively recovers obscured structural cues during live inference.

This study proposes a modular restoration pipeline that repairs occluded visual signals prior to classification. To test how well this works, we applied our method to the challenge of recognizing emotions from faces, using a well-known dataset called RAF-DB. Our three-step process was carefully evaluated in this context. A baseline ResNet-34 trained on clean data achieved 76.4% accuracy on clean inputs but dropped to 61.4% under occlusion. Standard occlusion-aware retraining improved occluded accuracy to 71.4%, establishing a tolerance-based ceiling. By integrating DeepFill v2, guided by dynamic facial landmark and texture-variance masking as an inference-time preprocessor, the proposed framework restored occluded regions and increased accuracy to 77.3%, effectively closing the clean-occluded domain gap. These findings indicate that active, inference-time signal reconstruction provides a superior approach for enhancing the robustness of vision classifiers against real-world obstructions without requiring end-to-end retraining.

Keywords—affective computing, facial emotion recognition, facial landmarks, generative adversarial networks, image inpainting, occlusion, ResNet-34

I. INTRODUCTION

Deep learning models encounter a generalisation bottleneck when transitioning from curated laboratory benchmarks to unconstrained, real-world environments. In controlled settings with frontal orientations, uniform

lighting, and unobstructed viewpoints, deep convolutional neural networks (CNNs) achieve near-human diagnostic accuracy in visual classification tasks. However, real-world vision pipelines deployed in uncontrolled environments face unpredictable physical corruptions, particularly partial occlusion from accessories, wearable items, hands, and environmental obstructions. As deep neural models learn rigid spatial patterns from clean benchmarks, localised visual obstructions distort intermediate feature representations, substantially reducing model reliability and classification confidence.

This vulnerability is particularly pronounced in affective computing, an interdisciplinary field formalised by Picard [1] to enable intelligent systems to interpret and adapt to human emotional states. Within affective computing, Facial Emotion Recognition (FER) serves as a critical perceptual component for applications such as clinical mental health monitoring, intelligent driver assistance, and interactive educational tools.

Although deep architectures like VGGNet, ResNet, and MobileNet have supplanted handcrafted descriptors (e.g., HOG, LBP) by learning structured features directly from image data [2], these models remain unreliable under unconstrained conditions. Affective signals are predominantly concentrated in fine-grained, localised facial structures around the periorbital (eyes, eyebrows) and perioral (mouth) regions. As Zhang et al. [3] show, standard residual networks degrade when even one of these critical expression-bearing sub-regions is obstructed.

Prior research addressing in-the-wild facial occlusion has predominantly converged on two typical paradigms:

- **Occlusion-Tolerant Classification:** This defensive strategy adapts the classifier to operate on corrupted or incomplete inputs through partial-face feature-learning, attention mechanisms that actively suppress noisy channels, or feature-masking augmentations during training [4], [5]. Although this paradigm boosts classification tolerance, it rests on a fundamental assumption: occluded regions are irrecoverable noise. Consequently, vital diagnostic information obscured beneath the obstruction, such as lip curvature beneath a face mask or eyebrow furrowing behind a hand, is permanently discarded rather than recovered.
- **Offline Generative Augmentation:** A parallel branch that employs Generative Adversarial Networks (GANs) with contextual attention techniques to learn global structural semantics and hallucinate missing pixels [6], [7]. Foundational inpainting frameworks introduced by Yu et al. [6] enable generators to borrow contextual information from visible areas, and coarse-to-fine generative variants have been explored to repair masked facial regions for motion analysis [7]. However, researchers have deployed generative methods in visual recognition almost exclusively as offline processing tools during training, either to balance skewed class distributions or to synthesise occluded variations. During deployment, the live inference pipeline stays passive, forcing classifiers to process corrupted inputs without real-time generative support.

To address the limitations of both defensive tolerance and offline augmentation, this study introduces an active, modular inference-time generative restoration framework. Rather than modifying downstream classifiers to discard missing information or retraining complex backbones, the proposed model adds a plug-and-play preprocessing stage that dynamically reconstructs occluded features at runtime before feature extraction.

The primary contributions of this work are summarized as follows:

- I. **Inference-Time Restoration Paradigm:** This work demonstrates that active, runtime generative reconstruction provides a higher performance ceiling compared to traditional occlusion-tolerant training by recovering latent signals rather than suppressing them.
- II. **Modular Pipeline Architecture:** An end-to-end inference framework that integrates MediaPipe landmark localization (Lugaresi et al. [15]), with patch-level texture-variance analysis to guide a

DeepFill v2 contextual-attention generator (Yu et al. [16]) for real-time restoration of expression-bearing facial regions.

- III. **Empirical Validation across Three Evaluative Paradigms:** Using an 80:20 partition on the RAF-DB dataset (Li et al. [18]) with an occluded-only test split, three progressive configurations are evaluated:
 - A. **Baseline Transfer Learning:** A standard ResNet-34 classifier (He et al. [17]) trained exclusively on clean data.
 - B. **Occlusion-Tolerant Benchmark:** An optimised classifier trained with synthetic occlusion pairings and equipped with a channel-attention weighting head, representative of tolerance-based approaches (Chen et al. [4]; Song et al. [11]).
 - C. **Proposed Generative Restoration Pipeline:** An active inference-time system combining MediaPipe perception (Lugaresi et al. [15]) and DeepFill v2 inpainting (Yu et al. [16]) with downstream classification.

The results demonstrate that the proposed restoration framework surpasses the tolerance-based benchmark, achieving a 6.1% accuracy improvement and nearly eliminating the performance disparity between clean and occluded samples.

II. LITERATURE REVIEW

Evolution of Facial Emotion Recognition: From Handcrafted Features to Deep CNNs.

Facial Emotion Recognition (FER) has undergone a fundamental paradigm shift from classical feature engineering to end-to-end deep learning pipelines. Early methodologies relied heavily on manually designed texture and geometric descriptors, including Local Binary Patterns (LBP), Histograms of Oriented Gradients (HOG), and Scale-Invariant Feature Transforms (SIFT). While these descriptors, classified using Support Vector Machines (SVMs), performed adequately on early constrained laboratory datasets such as CK+ and JAFFE, they lacked the generalisation capacity needed to accommodate in-the-wild variability in illumination, head pose, and facial morphology [8].

The adoption of Convolutional Neural Networks (CNNs) revolutionised visual recognition by enabling models to learn hierarchical spatial representations directly from raw pixel matrices [2]. Foundational architectures such as VGGNet, ResNet (He et al. [17]), and MobileNet have since established baseline performance across standard affective benchmarks. Transfer learning using deep networks pretrained on large-scale datasets (e.g.,

ImageNet) has become the standard mechanism to accelerate convergence and reduce overfitting on emotion-specific datasets [2].

However, standard CNN architectures degrade sharply in unconstrained, real-world deployments. While proprietary or specialised architectures such as NVIDIA’s EmotionNet [10] provide competitive benchmark results, they frequently require rigid execution environments, extensive parameter refinement, or specialised hardware ecosystems, limiting their modifiability as lightweight, plug-and-play solutions. Furthermore, recent explorations with modern architectures such as Vision Transformers [20] and advanced margin-based loss functions [21] emphasise that standard architectural scaling alone cannot completely resolve the distributional shift caused by uncontrollable physical corruptions in the wild.

The Occlusion Generalisation Bottleneck in Affective Computing

In real-world vision systems, partial facial occlusion, arising from medical masks, sunglasses, winter wear, hands, hair, or severe environmental shadows, forms a critical failure mode. Unlike general object recognition where semantic identity can often be inferred from coarse global shapes, affective computing relies on localised, fine-grained micro-expressions concentrated in the periorbital (eyebrows, eyes) and perioral (mouth, lips) zones.

When any of these expressive zones are obstructed, deep feature extractors experience severe feature distortion, resulting in misclassifications and collapsed prediction confidence [3], [9]. Zhang et al. [3] documented that standard classifiers routinely lose more than 15% accuracy when transitioning from clean to occluded samples.

Prior research on in-the-wild facial occlusion has largely followed two competing paradigms: *defensive occlusion tolerance* and *offline generative augmentation*.

Defensive Occlusion-Tolerant Strategies

The traditional approach to handling missing facial information focuses on training the classifier to tolerate corruption. One widespread technique is occlusion-aware training, where networks are exposed during optimisation to synthetic occlusions such as CutOut, random erasing, or artificial rectangular masks (Song et al. [11]). Although synthetic exposure forces the network to discover alternative spatial pathways, the model does not recover the lost diagnostic cues; it merely learns to ignore the occluded patches [11].

A parallel line of work incorporates attention models and partial-face representations to dynamically re-weight

feature maps [4], [5]. For example, the Region Attention Network (RAN) proposed by [22] explicitly computes weighting factors to prioritise visible facial sub-regions while limiting occluded channels. Similarly, Ke et al. [5] combined focal transformers with partial feature masking to reduce missing facial regions. While these tolerance-based strategies improve baseline robustness, they share a fundamental design limitation: they treat occluded regions as irrecoverable noise to be discarded. Crucial diagnostic signals masked beneath obstructions, such as subtle nasolabial folds or lip tremors, are permanently lost to the classification pipeline.

Generative Inpainting: Shifting from Training-Time Augmentation to Runtime Restoration

The second paradigm uses Generative Adversarial Networks (GANs) for image inpainting [12]. Rather than treating corrupted regions as noise, GANs formulate inpainting as a semantic completion task, hallucinating visually and structurally plausible content conditioned on surrounding uncorrupted context. Foundational generative frameworks, such as Context Encoders by [23], demonstrated that training networks to reconstruct lost regions forces the model to learn deep semantic representations.

Modern inpainting frameworks have greatly improved structural coherence. Yu et al. [6] introduced contextual attention layers that enable generators to search for and borrow relevant feature patches from distant, unoccluded regions of the image. EdgeConnect [13] further refined this through decoupling structural edge generation from texture synthesis, ensuring sharp boundary reconstruction across intricate anatomical contours like the eyes and mouth. In DeepFill v2, Yu et al. [16] replaced standard convolutional operators with learnable gated convolutions, allowing the network to process free-form, arbitrary mask geometries with great accuracy.

Despite the efficacy of generative inpainting, its deployment within affective computing and visual recognition has remained predominantly confined to offline data augmentation. In standard implementations, GANs are used before training to balance class distributions or synthesise large-scale occluded datasets [14]. Once deployed for live inference, however, the classification model runs without generative assistance, leaving it vulnerable to novel real-world corruptions.

A few recent studies have initiated the transition toward active restoration. Notably, Man and Cho [7] implemented a two-stage coarse-to-fine GAN to repair masked facial regions prior to emotion evaluation. Following this trajectory, our work addresses the in-the-wild robustness gap by formulating a decoupled, inference-time restoration pipeline. By pairing dynamic

runtime masking from MediaPipe 468-point facial mesh tracking [15] with patch-level texture variance and DeepFill v2 contextual inpainting [16], our approach restores occluded affective signals dynamically during inference, providing a modular, plug-and-play enhancement for standard classifiers without end-to-end retraining.

III. RESEARCH METHODOLOGY

The methodology utilises a progressive, comparative framework to assess whether active generative restoration at inference time yields measurable performance improvements beyond the limits of traditional defensive classifiers. By systematically isolating the transfer learning baseline and the occlusion-tolerant benchmark, observed performance gains in the final configuration can be attributed specifically to the generative restoration module rather than to downstream classifier adjustments.

Dataset Preparation and In-the-Wild Partitioning

All experiments were conducted on the Real-World Affective Faces Database [18]. RAF-DB contains varied facial expressions exhibiting unconstrained real-world variability in lighting, pose, and natural occlusions. To evaluate model resilience under in-the-wild corruptions, the dataset was partitioned using an 80:20 split:

- Training Partition (80%): Composed of clean images and synthetically occluded pairs used to optimise the classification backbones and tolerance baselines.
- Evaluation Partition (20%): Reserved strictly for occluded facial samples. Testing exclusively on obstructed faces directly stress-tests in-the-wild diagnostic performance, avoiding artificial score inflation from clean samples.

Progressive Evaluative Framework

Baseline Transfer Learning Classifier (He et al. [17]):

To quantify the fundamental weakness of unaugmented deep learning models under in-the-wild corruptions, a standard ResNet-34 architecture pretrained on ImageNet was fine-tuned exclusively on clean training images. Because standard residual networks learn rigid spatial representations from clean distributions, this baseline establishes the lower-bound diagnostic degradation when critical facial regions are occluded.

Occlusion-Tolerant Benchmark (Chen et al. [4]; Song et al. [11]; Wang et al. [22]):

To establish the performance ceiling achievable through defensive tolerance-based learning alone, we trained a second ResNet-34 model on a balanced 1:1 mixture of clean and synthetically occluded images. We replaced the standard output layer with a custom dual-layer, attention-aware classification head to learn dynamic feature weighting. This head computes channel-wise attention weights (α) to focus on uncorrupted spatial pathways:

$$\alpha = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot x)) \quad (1)$$

$$x_{\text{attended}} = \alpha \odot x \quad (2)$$

$$\hat{y} = W_{\text{cls}} \cdot x_{\text{attended}} \quad (3)$$

where W_1 and W_2 are learnable projection weights, σ denotes the sigmoid activation function, and $\odot$ represents the Hadamard element-wise product.

To penalise classification errors on obstructed samples, an Occlusion-Aware Cross-Entropy Loss

$$L_{\text{total}} = L_{\text{CE}} + \beta \cdot L_{\text{penalty}} \quad (4)$$

Where L_{CE} represents class-weighted cross-entropy, L_{penalty} is the explicit loss penalty for misclassified occluded samples, and β is a scaling hyperparameter.

Proposed Model: GAN-Augmented Inference-Time Restoration Pipeline:

The proposed model transitions from defensive noise suppression to proactive signal restoration. Rather than forcing the classifier to rely on partial faces, DeepFill v2 (Yu et al. [16]) is integrated as a decoupled, plug-and-play preprocessor during live inference.

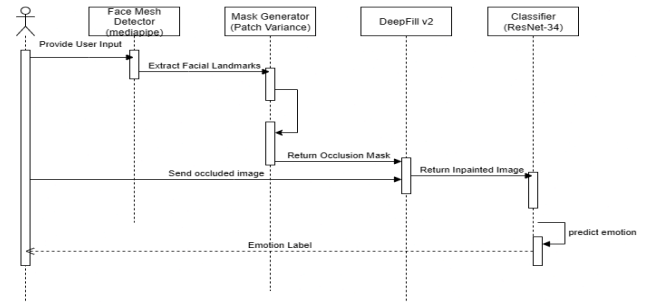


Fig. 1. Sequence Diagram of the Proposed GAN-Augmented Inference Pipeline.

Dynamic Inference-Time Restoration Logic

Unlike training-time augmentation, the proposed restoration pipeline operates in real time during live

deployment when ground-truth occlusion masks are unavailable. The framework dynamically detects and inpaints corruptions in three sequential steps:

- **Step 1: Facial Landmark Localisation:** The input image is processed through the MediaPipe perception pipeline [15], which extracts 468 high-density 3D facial landmarks to define the facial convex hull and isolate anatomical regions of interest.
- **Step 2: Dynamic Texture-Variance Mask Estimation:** Within the localized facial boundary, the image is divided into uniform $10 * 10$ pixel patches. Patches exhibiting local pixel variance below a predefined heuristic threshold, characteristic of uniform, flat textures like medical masks, sunglasses, or solid obstructions, are dynamically flagged to generate a binary occlusion mask.



Fig. 2. Landmark-Guided Dynamic Mask Generation. Illustration of the transition from (a) raw input to (b) 468-point facial mesh localization and (c) final patch-based variance mask estimation.

Source: Base image adapted from Pexels (available at: <https://www.pexels.com/photo/close-up-photo-of-a-woman-removing-her-blue-sleep-mask-6963176/>) and processed for the experimental pipeline..

- **Step 3: Contextual Inpainting and Classification:** Guided by the dynamically generated binary mask, the DeepFill v2 generator reconstructs the missing facial cues using gated convolutions and contextual attention layers that copy relevant structural details from uncorrupted facial areas [16]. The restored, complete facial representation is then passed to the downstream classifier for final emotion prediction.

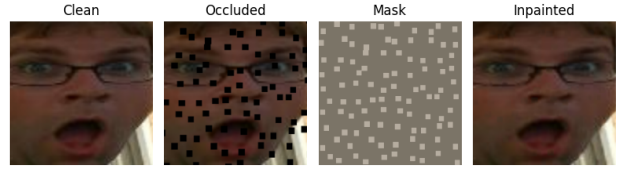


Fig. 3. Generative Image Restoration Pipeline. (a) Input image from the RAF-DB dataset; (b) Synthetic patches added to emulate occlusion; (c) Final patch-based variance mask estimation; (d) High-fidelity restoration using the DeepFill v2 contextual attention GAN.

Source: Samples extracted and processed from the Real-World Affective Faces Database (RAF-DB) [19].

Implementation and Optimisation Configuration

We implemented all classification models in PyTorch [19] and trained them on a GPU-accelerated workstation for over 50 epochs. Training optimisation used the Adam optimiser with a ReduceLROnPlateau scheduler (patience of 5 epochs) to maintain stability.

A tiered learning rate optimization strategy was employed to preserve low-level spatial filters while adapting high-level representations to complex occlusions:

- **Convolutional Backbone (Layer 4):** Assigned a fine-tuning learning rate of $3 * 10^{-4}$ to adapt high-level spatial representations to complex occlusion patterns.
- **Attention-Aware Head:** Optimized at a learning rate of $1 * 10^{-4}$ for stable, fine-grained convergence of the channel-weighting parameters.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

All experimental evaluations were conducted using PyTorch on a GPU-accelerated computing environment. Classification models followed a standardised 50-epoch training schedule, with the validation loss plateauing around epoch 45 across configurations.

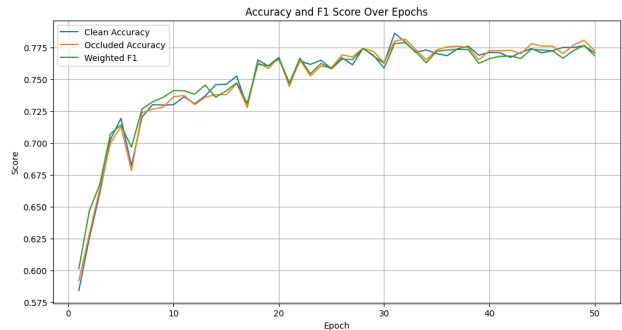


Fig. 4. Performance Convergence of the GAN-Augmented (Proposed Restoration) Pipeline. The plot illustrates the comparative accuracy

between clean and GAN-restored occluded samples across 50 training epochs.

Note the convergence of clean and restored accuracy, indicating the effectiveness of the inference-time inpainting in recovering emotional signals.

As illustrated in Fig. 4, the clean and GAN-restored occluded accuracy trajectories converged tightly throughout optimization, maintaining an average difference of less than 1% across epochs. At peak performance, the occluded test accuracy slightly exceeded clean accuracy by a 0.2% margin (77.3% vs. 77.1%). This indicates that inference-time contextual reconstruction not only recovers missing visual structures but can also clarify subtle expression cues that remain ambiguous or noisy in unoccluded in-the-wild input.

Comparative Performance Analysis across Paradigms

To validate the primary thesis, all three evaluative configurations were tested against the 20% occluded test partition of the RAF-DB dataset.

Model Evaluation Paradigm	Occluded Accuracy (%)	F1-Score
Baseline Transfer Learning Classifier (He et al. [17])(Trained on clean data)	61.4	0.56
Occlusion-Tolerant Benchmark (Chen et al. [4]; Wang et al. [22])(Trained on 1:1 clean/occluded mix)	71.4	0.70
Proposed Model: GAN-Augmented Inference-Time Restoration Pipeline	77.3	0.76

Table 1. Quantitative Performance Metrics on Occluded Test Samples across Experimental Paradigms.

The empirical results in Table 1 demonstrate a clear two-phase performance push: Baseline Fragility to Tolerance Ceiling (+10.0%):

- Exposing the baseline ResNet-34 classifier to occluded test samples resulted in severe degradation, dropping from 76.4% on clean data to 61.4% on occluded data. Training on a 1:1 clean/occluded mix combined with the attention-aware head (Occlusion-Tolerant Benchmark) improved occluded accuracy to 71.4% (F1-score: 0.70). This 10.0% gain establishes the theoretical ceiling of defensive noise-suppression techniques.
- Active Restoration Push (+5.9% / +6.1%): Integrating the DeepFill v2 inpainting module at inference time elevated occluded classification accuracy to 77.3% (F1-score: 0.76). This provides empirical confirmation that active, inference-time generative reconstruction outperforms defensive tolerance strategies by directly restoring diagnostic affective cues rather than treating occlusions as discarded noise.

Per-Class Performance and Error Analysis

To evaluate the class-level behavior of the proposed restoration pipeline, a detailed per-class F1-score and confusion matrix analysis was conducted on the occluded evaluation set.

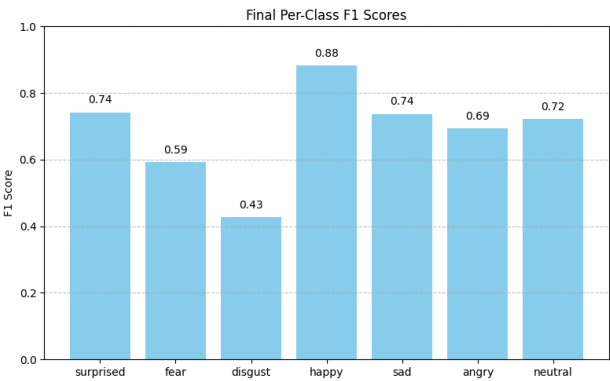


Fig. 5. Detailed F1-Score Performance across Emotion Categories for the Proposed Restoration Pipeline. The bar chart illustrates the model's reliability in classifying specific affective states following GAN-based restoration on occluded samples.

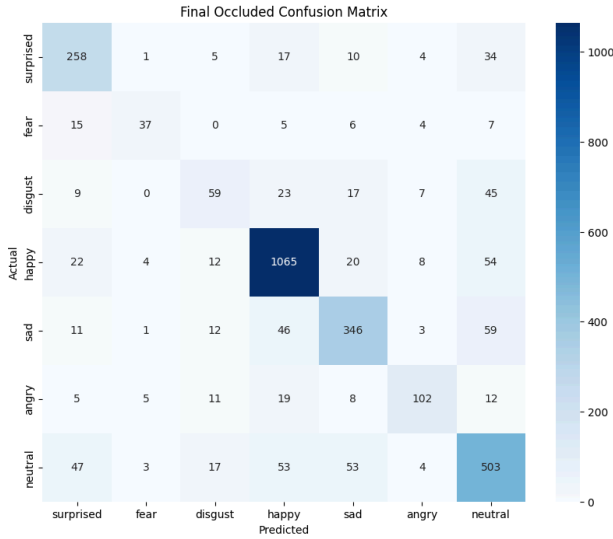


Fig. 6. Confusion Matrix of the GAN-Augmented FER Pipeline on the RAF-DB Occluded Test Set. Diagonal elements represent the precision of predictions for each emotional class, while off-diagonal elements illustrate inter-class misclassification patterns.

As shown in Fig. 5 and Fig. 6, the proposed pipeline achieved superior performance on expressive affective states:

- **High-Fidelity Classes:** Happiness (0.88 F1) and Surprise (0.74 F1) exhibited the highest classification fidelity. These expressions involve prominent geometric deformations (e.g., broad smiles, raised brows, open mouths) that provide strong structural priors for contextual GAN inpainting. Neutral expressions also demonstrated strong reliability (0.72 F1) due to consistent facial symmetry.
- **Challenging Fine-Grained Classes:** Subtler emotions, specifically Disgust (0.43 F1) and Fear (0.59 F1), yielded lower recognition rates. This behaviour is attributed to two factors: (i) the nuanced, micro-muscle movements characteristic of disgust (e.g., nose wrinkling) are more susceptible to generative smoothing during inpainting, and (ii) the inherent class imbalance of the RAF-DB dataset limits the representation of these categories during backbone training.

Consolidated Reliability Metrics

To ensure performance improvements were not biased toward majority classes in the imbalanced RAF-DB distribution, we aggregated macro-level evaluation metrics across all emotion categories.

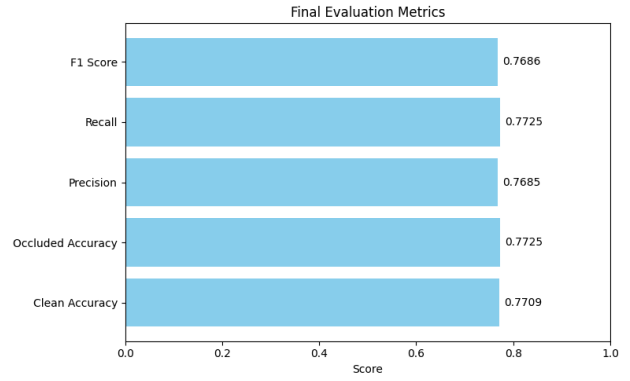


Fig. 7. Consolidated Evaluation Metrics for the GAN-Augmented (Proposed Restoration) Pipeline. The bar chart illustrates the equilibrium between Accuracy, Precision, Recall, and F1-Score, validating the model’s reliability on the imbalanced RAF-DB occluded test set.

The consolidated metrics (Accuracy: 0.7725, Recall: 0.7725, Precision: 0.7685, F1-Score: 0.7686) demonstrate balanced equilibrium across predictive dimensions. The tight grouping in the 76–77% band confirms that the proposed modular restoration pipeline hardens the classifier against in-the-wild occlusions by restoring expression-bearing regions prior to downstream inference.

V. CONCLUSION AND FUTURE WORK

This study investigated integrating Generative Adversarial Networks (GANs) into Facial Emotion Recognition (FER) pipelines to address diagnostic degradation caused by unconstrained facial occlusion in real-world environments. By shifting from a defensive occlusion-tolerance paradigm to an active inference-time generative restoration paradigm, we developed and empirically validated a modular, decoupled framework. The experimental findings confirm that active runtime restoration provides a higher performance ceiling than conventional training-time robustness techniques.

While the occlusion-tolerant benchmark achieved a 71.4% accuracy ceiling through synthetic data augmentation and attention-aware channel weighting, integrating the DeepFill v2 inpainting engine as a live preprocessor elevated occluded classification accuracy to 77.3% on the RAF-DB benchmark. By dynamically detecting occluded regions via MediaPipe landmark tracking and texture variance analysis, the proposed pipeline effectively recovered missing affective signals and periorbital micro-expressions, allowing downstream classifiers to operate on complete visual representations without requiring end-to-end retraining.

Despite the empirical gains achieved by the proposed pipeline, several technical limitations remain to be addressed:

- **Semantic Emotion Fidelity:** Pretrained DeepFill v2 weights are optimised for generic visual naturalness rather than emotional preservation. Consequently, generative reconstruction may occasionally introduce subtle semantic drift that deviates from the subject's ground-truth emotional state.
- **Heuristic Masking Sensitivity:** The reliance on patch-level texture variance for dynamic mask estimation can occasionally lead to false positives, misinterpreting environmental shadows or uniform skin regions as physical occlusions.
- **Computational Latency:** Integrating a generative model into the live inference pipeline introduces additional computational overhead. While viable for real-time execution on GPU-accelerated workstations, constrained edge devices require further latency optimisation.
- **Dataset Distribution Skew:** Inherent class imbalances in naturalistic datasets like RAF-DB continue to constrain performance on minor, nuanced affective states such as Fear and Disgust.

To build upon the foundation established in this work, future research should explore the following directions:

- **Emotion-Preserving Generative Objectives:** Designing domain-specific GAN or diffusion loss functions conditioned on facial action units (AUs) to enforce strict emotional consistency during inpainting.
- **Semantic Occlusion Segmentation:** Replacing heuristic variance masking with lightweight, learned segmentation networks trained to dynamically detect diverse real-world occlusions with pixel-level precision.
- **Advanced Backbone Integration:** Evaluating the plug-and-play restoration module with modern Vision Transformer (ViT) and EfficientNet backbones to capture richer global contextual dependencies.
- **Edge-Device Optimisation:** Applying model compression techniques, such as knowledge distillation and structured pruning, to the generative inpainting engine to enable efficient, low-latency deployment in mobile and interactive embedded systems.

CONFLICT OF INTEREST

The author declares that there is no conflict of interest regarding the publication of this paper.

AUTHOR CONTRIBUTIONS

MN developed the model's algorithm; MN wrote and documented the manuscript; KO did the final editing and correction; all authors approved the final version.

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